

Homework Assignment 1: Slab-Geometry Boltzmann Equation

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1 Question 1 - Useful identity for the solution of the time-dependent slab-geometry Boltzmann equation

In this question we are asked to prove the following identity:

$$\psi(x, \mu, t; c) = ce^{-(1-c)v\Sigma_t t}\psi(cx, \mu, ct; 1) \quad (1)$$

(the exponent is written as $-(1-c)t$ in the units $v = \Sigma_t = 1$). We will do it in two parts. First, we will write down the equation that $\tilde{\psi} = \psi(cx, \mu, ct; 1)$ satisfies, and then we will use it in order to prove that the product $ce^{-(1-c)v\Sigma_t t}\tilde{\psi}$ satisfies the same equation as $\psi(x, \mu, t; c)$.

1.1 The equation for $\tilde{\psi}$

Notation: We will denote $\frac{\partial \tilde{\psi}(x, \mu, t)}{\partial \tilde{x}}$ as the full derivative (before applying the chain rule) and $\frac{\partial \tilde{\psi}(x, \mu, t)}{\partial x}$ as the partial derivative (after applying the chain rule, and assuming that the cx and ct are parameters *substituted* into the derivative (and not those that the derivative is taken with respect to)). We will denote the same for the time derivative. First, from the definition of $\tilde{\psi}$ we have:

$$\frac{1}{v} \frac{\partial \tilde{\psi}}{\partial \tilde{t}} + \mu \frac{\partial \tilde{\psi}}{\partial \tilde{x}} + \Sigma_t \tilde{\psi} = \frac{\Sigma_t}{2} \int_{-1}^1 \tilde{\psi} d\mu' \quad (2)$$

which after applying the chain rule, $\frac{\partial \tilde{\psi}}{\partial \tilde{t}} = \frac{1}{c} \frac{\partial \tilde{\psi}}{\partial t}$ and the same for x , and multiplying by c , becomes:

$$\frac{1}{v} \frac{\partial \tilde{\psi}}{\partial t} + \mu \frac{\partial \tilde{\psi}}{\partial x} + c\Sigma_t \tilde{\psi} = \frac{c\Sigma_t}{2} \int_{-1}^1 \tilde{\psi} d\mu' \quad (3)$$

So $\tilde{\psi}$ obeys the same equation as $\psi(x, \mu, t; c)$ except that it is removed at the rate $c\Sigma_t$ instead of Σ_t , and the exponential factor is what makes up this difference.

1.2 Proving the identity

In this part we are going to simply substitute the RHS of 1 into the LHS of the equation for $\psi(x, \mu, t; c)$ and show that it satisfies the same equation. After substitution and application of the chain rule we

get:

$$\text{LHS} = \frac{1}{v} \left[c(c-1)v\Sigma_t e^{-(1-c)v\Sigma_t t} \tilde{\psi} + c e^{-(1-c)v\Sigma_t t} \frac{\partial \tilde{\psi}}{\partial t} \right] + \mu c e^{-(1-c)v\Sigma_t t} \frac{\partial \tilde{\psi}}{\partial x} + \Sigma_t c e^{-(1-c)v\Sigma_t t} \tilde{\psi} \quad (4)$$

$$= c e^{-(1-c)v\Sigma_t t} \left[\frac{1}{v} \frac{\partial \tilde{\psi}}{\partial t} + \mu \frac{\partial \tilde{\psi}}{\partial x} + \Sigma_t \tilde{\psi} + (c-1)\Sigma_t \tilde{\psi} \right] \quad (5)$$

$$= c e^{-(1-c)v\Sigma_t t} \left[\frac{1}{v} \frac{\partial \tilde{\psi}}{\partial t} + \mu \frac{\partial \tilde{\psi}}{\partial x} + c\Sigma_t \tilde{\psi} \right] \quad (6)$$

$$= c e^{-(1-c)v\Sigma_t t} \frac{c\Sigma_t}{2} \int_{-1}^1 \tilde{\psi} d\mu' = \frac{c\Sigma_t}{2} \int_{-1}^1 c e^{-(1-c)v\Sigma_t t} \tilde{\psi} d\mu' \quad (7)$$

where 3 was used in the last line, and this is exactly the RHS of the equation for $\psi(x, \mu, t; c)$. The prefactor c itself does not follow from the equation, which is homogeneous, but from the initial condition: for the plane pulse $\psi(x, \mu, 0^+) = \frac{v}{2}\delta(x)$ for every c , and indeed $c \cdot \frac{v}{2}\delta(cx) = \frac{v}{2}\delta(x)$.

2 Question 2 - Derivation of the time-dependent planar scalar flux for a plane source

The relation between a plane source and a point source as studied in class is given by:

$$\phi_{\text{pl}}(x) = 2\pi \int_{|x|}^{\infty} \phi_{\text{pt}}(r) r dr \quad (8)$$

The solution taken from J. C. J. Paasschens is the point-source scalar flux $\phi_{\text{pt}}(r, t)$ for $c = 1$, and therefore the planar-source scalar flux is given by the integration given above.

$$\phi_{\text{pl}}(x, t) = 2\pi \int_{|x|}^{\infty} \frac{e^{-v\Sigma_t t}}{4\pi r^2} \delta(r - vt) r dr + 2\pi \int_{|x|}^{\infty} \frac{(1 - r^2/v^2 t^2)^{1/8}}{(4\pi vt/(3\Sigma_t))^{3/2}} e^{-v\Sigma_t t} G\left(v\Sigma_t t \left[1 - \frac{r^2}{v^2 t^2}\right]^{3/4}\right) \Theta(vt - r) r dr \quad (9)$$

$$= \frac{e^{-v\Sigma_t t}}{2vt} \Theta(vt - |x|) + \Theta(vt - |x|) \frac{2\pi e^{-v\Sigma_t t}}{(4\pi vt/(3\Sigma_t))^{3/2}} \int_{|x|}^{vt} \left(1 - \frac{r^2}{v^2 t^2}\right)^{1/8} G\left(v\Sigma_t t \left[1 - \frac{r^2}{v^2 t^2}\right]^{3/4}\right) r dr \quad (10)$$

$$= \frac{e^{-v\Sigma_t t}}{2vt} \Theta(vt - |x|) + \Theta(vt - |x|) \frac{2\pi e^{-v\Sigma_t t} (vt)^{1/2}}{(4\pi/(3\Sigma_t))^{3/2}} \int_{\frac{|x|}{vt}}^1 (1 - \xi^2)^{1/8} G\left(v\Sigma_t t [1 - \xi^2]^{3/4}\right) \xi d\xi \quad (11)$$

Note that the collided term carries $\Theta(vt - |x|)$ and not $\text{sign}(vt - |x|)$, since for $|x| > vt$ the factor $\Theta(vt - r)$ kills the integrand for every $r \geq |x|$ and the term vanishes.

The remaining integral is in fact elementary. Writing $a = v\Sigma_t t$ and substituting first $u = 1 - \xi^2$ and then $w = au^{3/4}$, the powers combine as $\frac{1}{6} + \frac{1}{3} = \frac{1}{2}$ and

$$\int_{\frac{|x|}{vt}}^1 (1 - \xi^2)^{1/8} G\left(a [1 - \xi^2]^{3/4}\right) \xi d\xi = \frac{1}{2} \int_0^{u_0} u^{1/8} G\left(au^{3/4}\right) du = \frac{2}{3a^{3/2}} \int_0^{w_0} \sqrt{w} G(w) dw \quad (12)$$

with $u_0 = 1 - x^2/(vt)^2$ and $w_0 = a(1 - x^2/(vt)^2)^{3/4}$. This also removes the integrable $(vt - r)^{-1/4}$ singularity at the front. The prefactor then collapses, $2\pi(vt)^{1/2} (3\Sigma_t/4\pi)^{3/2} \frac{2}{3a^{3/2}} = \frac{1}{2vt} \sqrt{\frac{3}{\pi}}$, so that Σ_t cancels and both terms share the same factor:

$$\phi_{\text{pl}}(x, t) = \frac{e^{-v\Sigma_t t}}{2vt} \left[1 + \sqrt{\frac{3}{\pi}} \int_0^{w_0} \sqrt{w} G(w) dw \right] \Theta(vt - |x|) \quad (13)$$

Finally, with the interpolation formula of Paasschens, $G(w) = e^w \sqrt{1 + b/w}$ with $b = 2.026$, the $\sqrt{w}$ rationalises the square root, $\sqrt{w}G(w) = e^w \sqrt{w + b}$, and the integral has a closed form:

$$\int_0^{w_0} e^w \sqrt{w + b} dw = e^{w_0} \sqrt{w_0 + b} - \sqrt{b} - \frac{\sqrt{\pi}}{2} e^{-b} \left[\text{erfi}(\sqrt{w_0 + b}) - \text{erfi}(\sqrt{b}) \right] \quad (14)$$

For evaluation it is convenient to use $\text{erfi}(z) = \frac{2}{\sqrt{\pi}} e^{z^2} D(z)$ with D the Dawson function, which turns the bracket into $e^{w_0} [\sqrt{w_0 + b} - D(\sqrt{w_0 + b})] - [\sqrt{b} - D(\sqrt{b})]$; since $w_0 \leq a$, the e^{w_0} is cancelled by the e^{-a} in front and nothing overflows. As a check, at the front $|x| \rightarrow vt$ we have $w_0 \rightarrow 0$ and the bracket vanishes, leaving the uncollided term alone.

Of course the relation proved in Q1 can be used to get the solution for $c \neq 1$ using a simple scaling and multiplication by an exponential factor of the solution for $c = 1$.